UNIVERSITI MALAYSIA PERLIS

QMQ27103 ENGINEERING STATISTICS
2026/2027 | SEMESTER 1

Redhwan Ali Hasan Abdullah
241090104-5

COMPARISON OF BOTTLED WATER PRICES BY BOTTLE SIZE AND BRAND IN MALAYSIA



1 BACKGROUND

Bottled water is widely consumed in Malaysia due to its convenience, variety of brands, and different bottle sizes. Prices may vary across brands, sizes, and online stores, and understanding these differences can help consumers make more informed and cost-effective choices.

2 OBJECTIVES

1. To compare the number of product listings for each brand.
2. To examine how the price per litre varies among the top 5 and bottom 5 products.
3. To compare the distribution of price per litre among different online stores.
4. To analyze the overall distribution of bottled water prices per litre.

3 DATA AND METHODS

Data Collection

A total of 150 bottled water products were collected from five online stores in Malaysia.

Variables

- Brand (qualitative)
- Bottle size (quantitative, mL)
- Online store (qualitative)
- Price per litre (quantitative, RM/L)

Method

- Graphical and numerical analysis using Excel.
- Bar chart, scatter plot, box and whisker plot, and histogram.

Purpose

- To visualize patterns and differences in bottled water prices.
- To support data-driven conclusions about brand, size, and store effects.

4 RESULTS AND DISCUSSION

4.1 Number of Product Listings by Brand

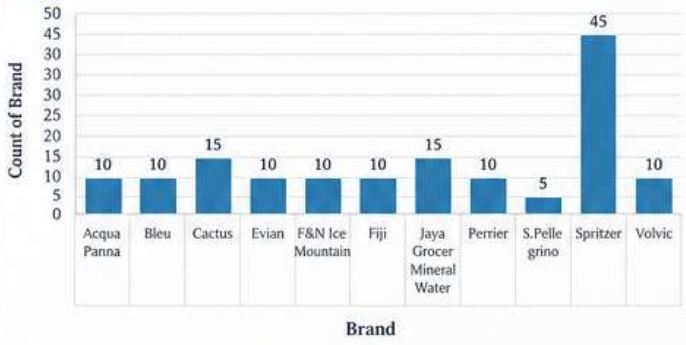


Figure 1. Number of product listings by brand

The bar chart shows the number of product listings for each brand. Spritzer has the highest number of listings (45), followed by Cactus and Jaya Grocer Mineral Water (15 each). S.Pellegrino has the lowest number of listings (5), while the other brands have 10 listings each.

4.2 Bottle Size vs Price per Litre: Top 5 and Bottom 5 Products

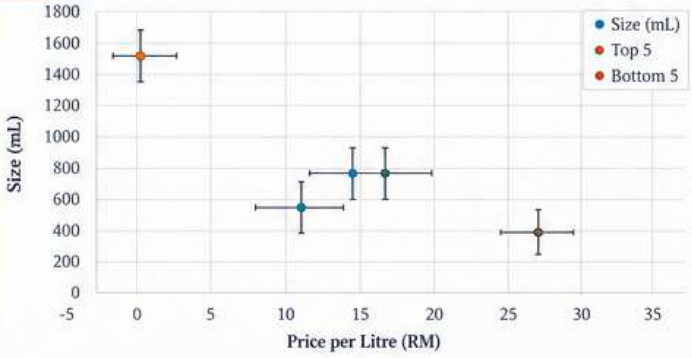


Figure 2. Bottle size vs price per litre for top 5 and bottom 5 products

The scatter plot compares the top 5 highest and bottom 5 lowest price-per-litre products. The lowest price-per-litre products generally have larger bottle sizes, while the highest price-per-litre products tend to have smaller bottle sizes.

4.3 Comparison of Price per Litre by Store

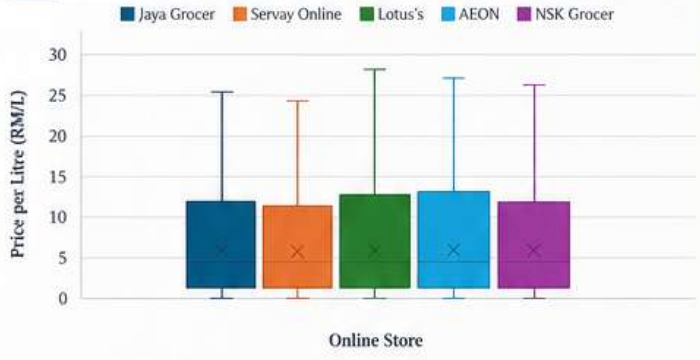


Figure 3. Comparison of price per litre by store

The box and whisker plot shows the distribution of price per litre for each online store. All stores have a wide range of prices, with similar medians and interquartile ranges. Lotus's and AEON show slightly higher maximum values compared to the other stores, indicating more high-priced products.

4.4 Distribution of Bottled Water Prices per Litre

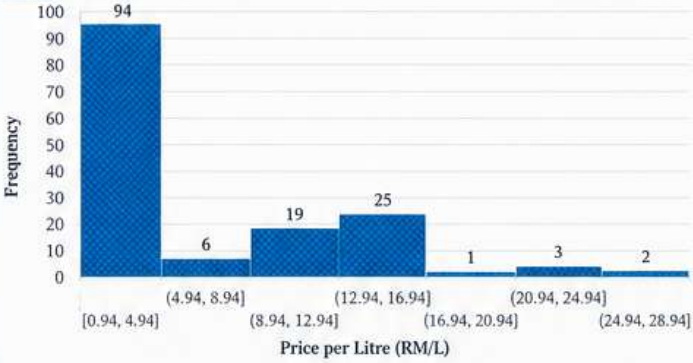


Figure 4. Distribution of bottled water prices per litre

The histogram shows the frequency distribution of price per litre. Most products (94 listings) have a price between RM 0.94 and RM 4.94 per litre. The frequency decreases as the price increases, with only a few products having prices above RM 16.94 per litre.

5 CONCLUSION

Based on the analysis of 150 bottled water products, the price per litre in Malaysia varies across brands, bottle sizes, and online stores. Most products are priced between RM 0.94 and RM 4.94 per litre, and larger bottle sizes generally offer lower prices per litre. While some brands such as Spritzer have more product listings, the price distributions across online stores are relatively similar. Overall, consumers can consider both bottle size and brand when choosing bottled water to get better value.

Key Findings

- Spritzer has the highest number of product listings.
- Most products are priced between RM 0.94 and RM 4.94 per litre.
- Larger bottle sizes tend to have lower prices per litre.
- All online stores show similar price distributions.
- There are a few high-priced products above RM 16.94 per litre.